Linear Algebra

Question 1

$\vec{\psi}_0 \cdot \vec{\psi}_1 = 0$ orthogonal but not orthonormal

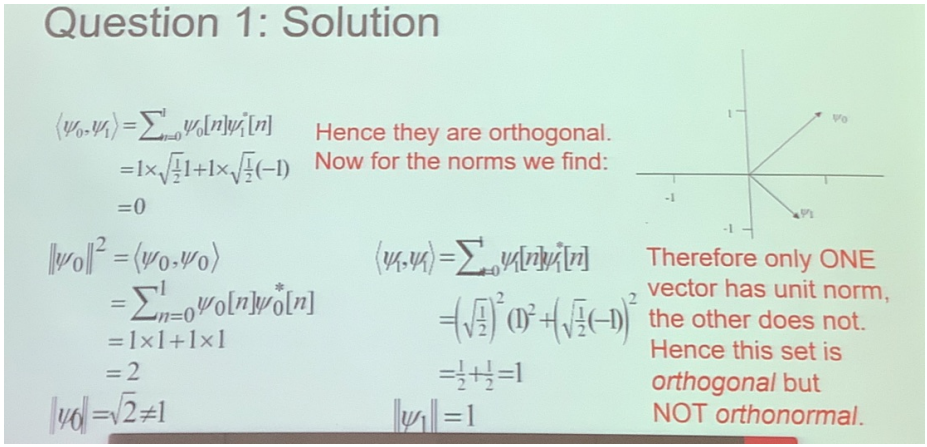
$\|\psi_0\| = \sqrt{2}$

$\|\psi_1\| = 1$

Consider the pair of vectors:

$$\psi_0 = (1,1) \qquad \psi_1 = \sqrt{\tfrac{1}{2}}(1,-1)$$

By calculating relevant inner products and norms, identify whether or not these vectors form an orthogonal ✓ or an orthonormal ✗ set.
Sketch these two vectors on a diagram to confirm your answer



$\langle \Psi_{11}, \Psi_{21} \rangle = 2a = 1$

$\langle \Psi_{11}, \Psi_{22} \rangle = 0$

Question 2

$\langle \Psi_{12}, \Psi_{21} \rangle = a^2 - \frac{2}{8}$

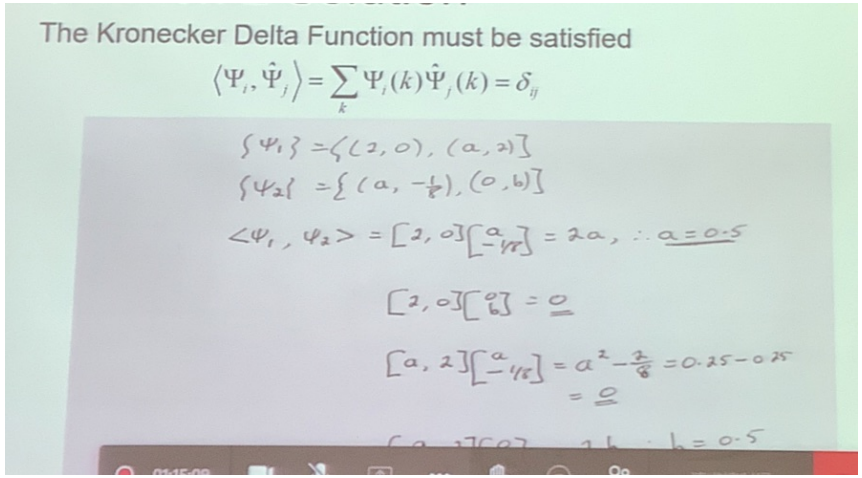
$\langle \Psi_{12}, \Psi_{22} \rangle = 2b = 1$

Two sets of functions are given by:

$\{\Psi_1\}=[(2,0),(a,2)]$

$\{\Psi_2\}=[(a,-1/8),(0,b)]$

State the condition required for these two sets to be a Dual Basis and determine the corresponding values of a and b.

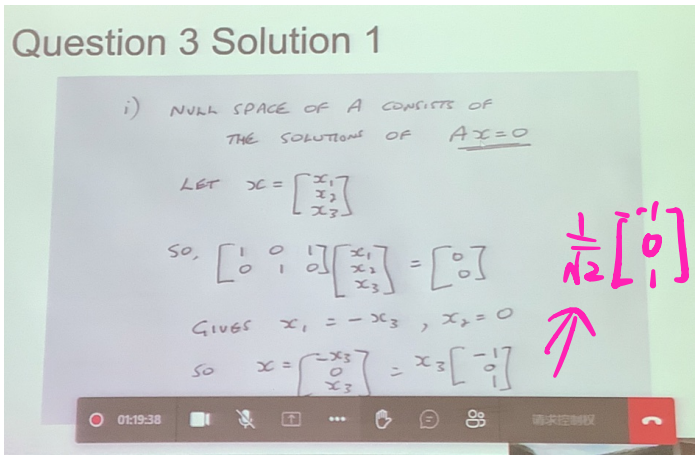


Question 3

The matrix A is:

$$A=\begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 0\end{bmatrix}$$

- Find an orthonormal basis of the NULL space of A.
- Find the RANK of A.
- Find an orthonormal basis of the ROW space of A.



Question 5 Part i)

$$v_1 = \frac{1}{3\sqrt{2}}\begin{bmatrix}1\\1\\-4\end{bmatrix},\ v_2 = \frac{1}{3}\begin{bmatrix}2\\2\\1\end{bmatrix},\ v_3 = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\\0\end{bmatrix}$$

Show that these three vectors are an orthonormal basis for R^3

$v_1 \cdot v_2 = 0$

$v_1 \cdot v_3 = 0$

$v_2 \cdot v_3 = 0$

orthogonal V

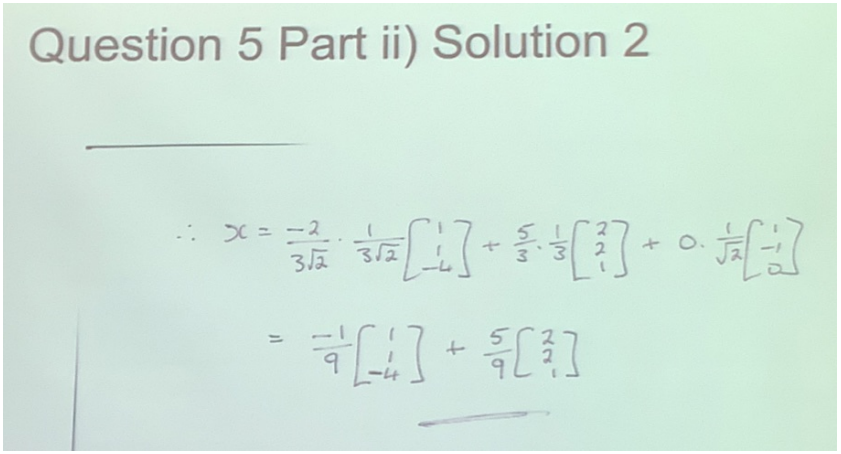
orthonormal V

$\|v_1\| = 1$, $\|v_2\| = 1$, $\|v_3\| = 1$

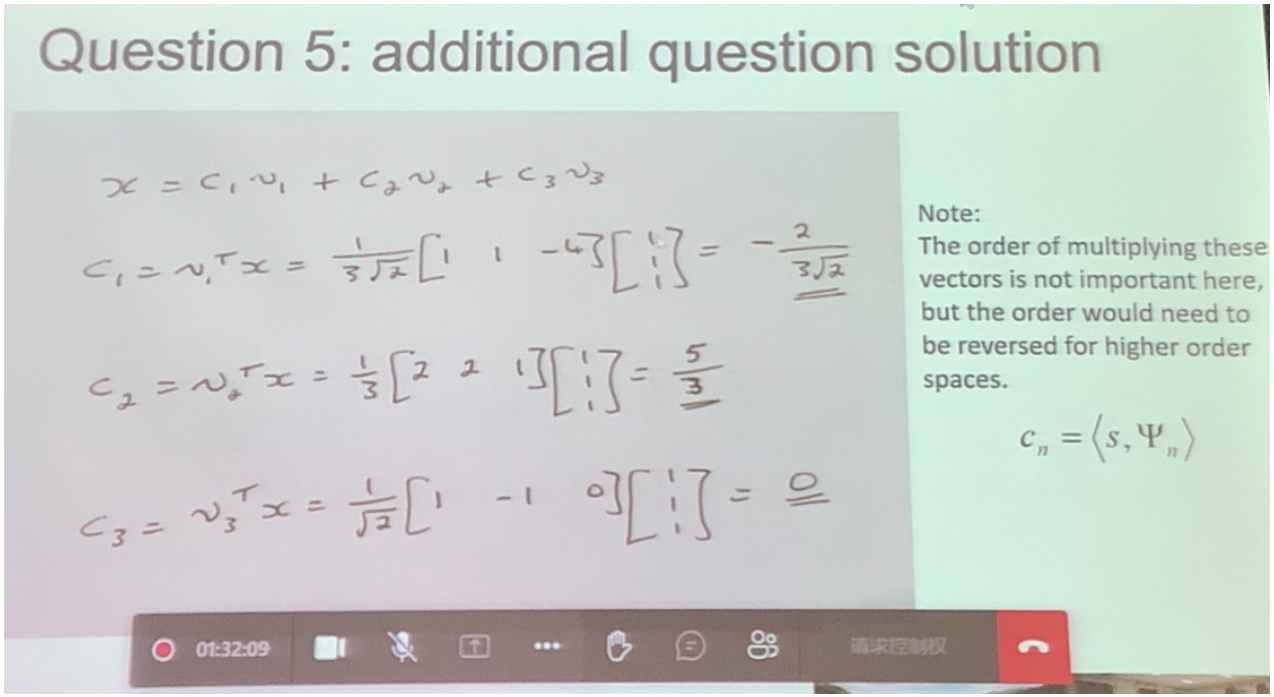
$c_n = v_n^T \cdot x$

Question 5 Part ii)

Vector $x = [1 \quad 1 \quad 1]'$



Write x as a linear combination of v1, v2 and v3.



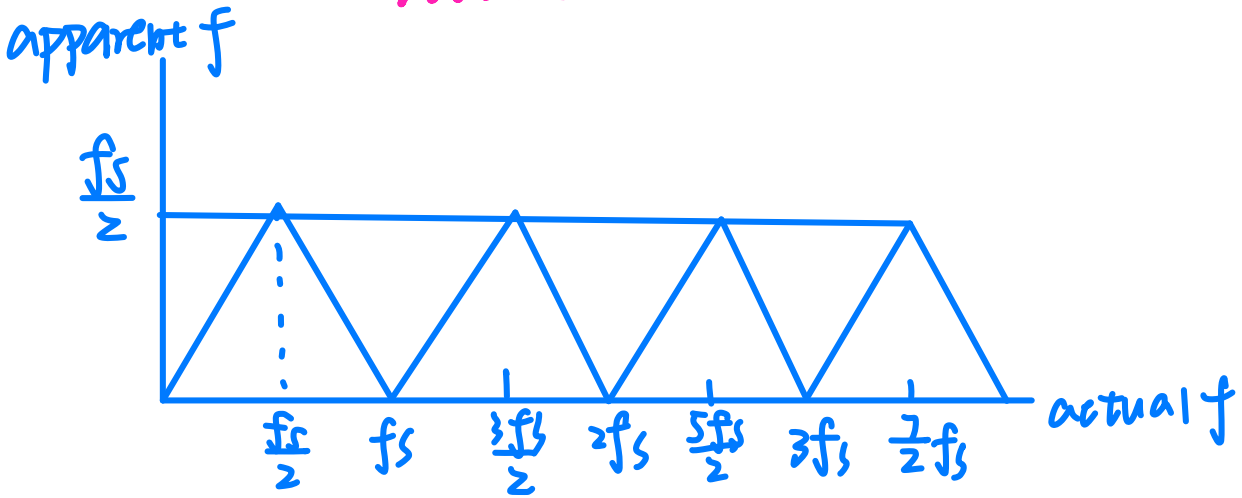
Sampling

Question 1

With the aid of a diagram, demonstrate the effect of the folding of an aliased signal.

For a sampling frequency of 4000 samples/second, what is the apparent frequency resulting from sampling signals of respectively 1000 Hz, 5000 Hz, 7500 Hz and 11,500 Hz?

1000Hz 1000Hz 500Hz 500Hz

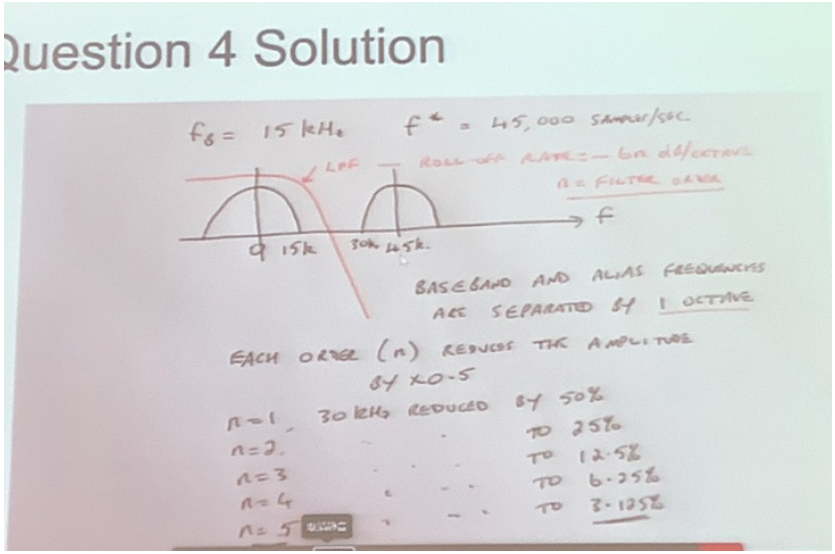


Question 4

Suppose we have a signal of bandwidth 15kHz that is sampled at a rate of 45,000 samples/second.

Sketch a portion of the frequency spectrum of the sampled signal showing the baseband spectrum and the first upper sideband.

Estimate the order of low-pass filter required to reduce the amplitude of alias frequencies to less than 5% of the baseband amplitudes.



octave

Question 5

Suppose we have a recording of a piece of music of duration 5 minutes.

We wish to obtain a DFT of the music.

Assume the quality of the recording is high with a bandwidth of 15kHz.

What is the minimum sampling rate? 30 kHz samples / second

What length of sequence is obtained from this sampling rate?

How could the implementation time of the DFT be reduced, and what would be the effect on the accuracy of the output?

DFT

mydft(xn,N)

- This can be written as:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}nk} = \sum_{n=0}^{N-1} x[n]\, W_N^{nk}$$

where $W_N = e^{-\frac{j2\pi}{N}}$



- The matrix formula for an N-point DFT can be written as $X = F_N x$
  - where x is the N-point input sequence of samples of a continuous signal
  - F_N is the N-by-N DFT matrix
  - X is the DFT of the signal

Discrete Fourier Transform - Matrix

➢ The N-by-N Fourier Matrix (N = 0…..(n-1)): *Count from 0

$$F_N = \begin{bmatrix} 1 & 1 & 1 \dots & 1 \\ 1 & \omega & \omega^2 \dots & \omega^{N-1} \\ 1 & \omega^2 & \omega^4 \dots & \omega^{2(N-1)} \\ \dots & \dots & \dots & \dots \\ 1 & \omega^{N-1} & \omega^{2(N-1)} \dots & \omega^{(N-1)^2} \end{bmatrix}$$

DFT:

➢ $W_N = e^{-\frac{j2\pi}{N}} = \omega$

➢ $F_N[k,n] = W_N{}^{nk} = \omega^{nk}$

Discrete Fourier Transform – 2x2 Matrix

- Now, consider $\omega = e^{-\frac{j2\pi}{N}} = [\cos\left(\frac{2\pi}{N}\right) - j\sin(\frac{2\pi}{N})]$ so $\omega^N = e^{-j2\pi} = 1$
- All the entries of F_N are on the unit circle in the complex plane and raising each one to the Nth power gives 1.
- For example, if N = 2, $e^{-j2\pi/2} = \cos(\pi) - j\sin(\pi)$ = -1
- Then $F_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, normalised.

Discrete Fourier Transform – 4x4 Matrix

- For N = 4 (N = 0…..3), $\omega = e^{-j2\pi/4} = [\cos\left(\frac{\pi}{2}\right) - i\sin(\frac{\pi}{2})] = -i$

- So the normalised 4x4 Fourier Matrix is:

- $F_4 = \frac{1}{\sqrt{4}}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & (-i)^2 & (-i)^3 \\ 1 & (-i)^2 & (-i)^4 & (-i)^6 \\ 1 & (-i)^3 & (-i)^6 & (-i)^9 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix}$

normalized

- Each row corresponds to an increasing frequency.

FFT

Question 3

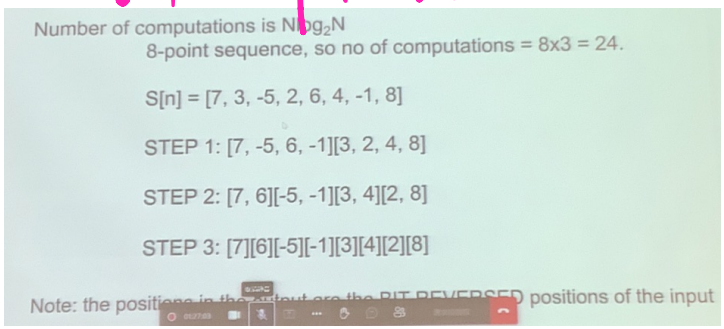
An FFT is a fast algorithm for implementing a DFT.

Estimate the approximate number of computations that are required to perform the FFT of an 8-point sequence.

One FFT structure is radix-2 decimation-in-time. Illustrate this FFT structure using the following 8-point sequence:

S[n] = [7, 3, -5, 2, 6, 4, -1, 8]

0 4 2 6 | 1 5 3 7



DCT

dct(xn)

1-Dimensional DCT

$$DCT[k] = c(k)\sum_{n=0}^{N-1} s[n]\cos\frac{\pi(2n+1)k}{2N} \qquad k = 0,1,2...N-1$$

$$DCT[k] = \langle s, \psi_k \rangle$$

$$c(k) = \begin{cases} \sqrt{1/N} & k = 0 \\ \sqrt{2/N} & k \neq 0 \end{cases}$$

c(k) is the normalisation factor.

- Orthonormal

$$\langle \psi_m, \psi_n \rangle = \begin{cases} 1 & \text{if } m = n \\ 0 & \text{if } m \neq n \end{cases}$$

The Basis Functions Ψ_k are the cosine terms in the definition.
They are calculated for each value of k, with n = 0….N-1

DCT Question 1

If N = 4 (input is a 4-point sequence)

For each value of k = 0….N-1, insert n = 0….N-1:

$$\psi_0 = (1,1,1,1)/2$$

$$\psi_1 = \sqrt{1/2}(\cos(\pi/8), \cos(3\pi/8), \cos(5\pi/8), \cos(7\pi/8))$$

$$\psi_2 = \sqrt{1/2}(\cos(\pi/4), \cos(3\pi/4), \cos(5\pi/4), \cos(7\pi/4))$$

$$\psi_3 = \sqrt{1/2}(\cos(3\pi/8), \cos(9\pi/8), \cos(15\pi/8), \cos(5\pi/8))$$

$$DCT[0] = \frac{1}{\sqrt{N}} \sum_{n=0}^{3} s[n] \qquad DCT[1] = \sqrt{\frac{2}{N}} \sum_{n=0}^{3} s[n] \cos\frac{\pi(2n+1)}{8}$$

$$DCT[2] = \sqrt{\frac{2}{N}} \sum_{n=0}^{3} s[n] \cos\frac{\pi(2n+1)}{4} \qquad DCT[3] = \sqrt{\frac{2}{N}} \sum_{n=0}^{3} s[n] \cos\frac{\pi(2n+1)3}{8}$$

4x4 DCT Basis Matrix

$$\Psi = \begin{bmatrix} 0.50 & 0.50 & 0.50 & 0.50 \\ 0.65 & 0.27 & -0.27 & -0.65 \\ 0.50 & -0.50 & -0.50 & 0.50 \\ 0.27 & -0.65 & 0.65 & -0.27 \end{bmatrix}$$

3x3 DCT Matrix

$$\Psi = \begin{bmatrix} 0.577x1 & 0.577x1 & 0.577x1 \\ 0.816x0.866 & 0.816x0 & 0.816x(-0.866) \\ 0.816x0.5 & 0.816x(-1) & 0.816x0.5 \end{bmatrix}$$

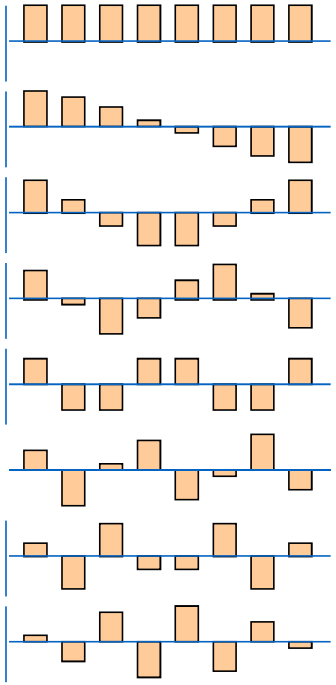
$$\Psi = \begin{bmatrix} 0.577 & 0.577 & 0.577 \\ 0.707 & 0 & -0.707 \\ 0.408 & -0.816 & 0.408 \end{bmatrix}$$

8x8 DCT Matrix

$$\Psi = \frac{1}{2}\begin{bmatrix} .71 & .71 & .71 & .71 & .71 & .71 & .71 & .71 \\ .98 & .83 & .56 & .19 & -.19 & -.56 & -.83 & -.98 \\ .92 & .38 & -.38 & -.92 & -.92 & -.38 & .38 & .92 \\ .83 & -.19 & -.98 & -.56 & .56 & .98 & .19 & -.83 \\ .71 & -.71 & -.71 & .71 & .71 & -.71 & -.71 & .71 \\ .56 & -.98 & .19 & .83 & -.83 & -.19 & .98 & -.56 \\ .38 & -.92 & .92 & -.38 & -.38 & .92 & -.92 & .38 \\ .19 & -.56 & .83 & -.98 & .98 & -.83 & .56 & -.19 \end{bmatrix}$$

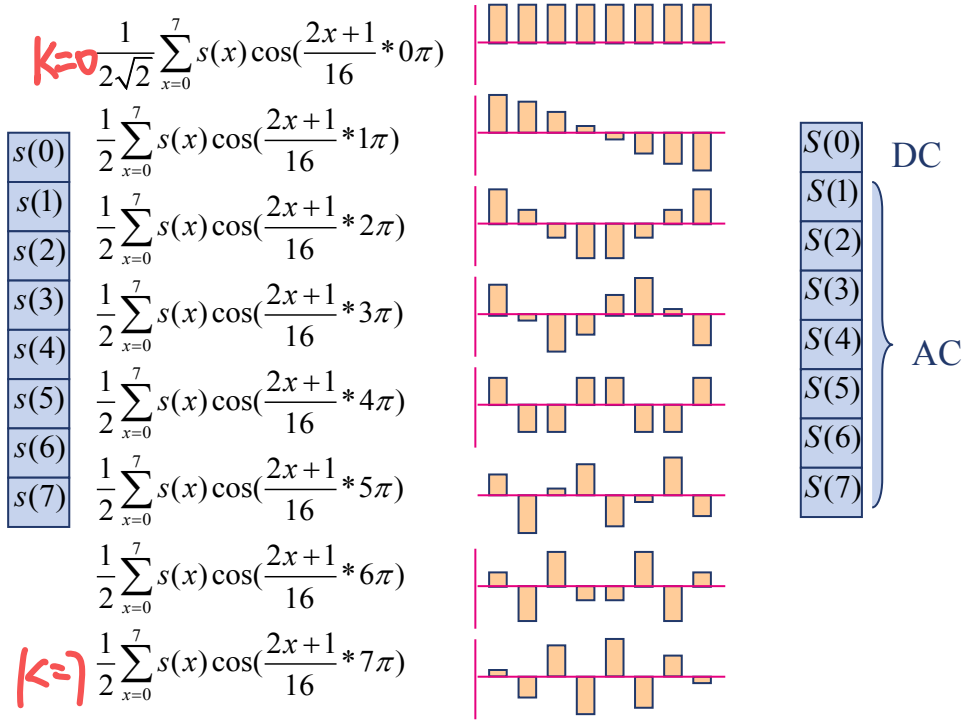
Comment

$$Output = \begin{bmatrix} 2.13 \\ -1.825 \\ 1.84 \\ 1.775 \\ 13.49 \\ 0.175 \\ 0.76 \\ 0.875 \end{bmatrix}$$



Input sequence data varies like this frequency

Discrete Cosine Transform – Basis Functions



Discrete Cosine Transform – Tutorial

8. Based on the given DCT output, explain what it implies on the input sequence.

$$\begin{bmatrix} 2.1 \\ 0.6 \\ -1.2 \\ 1.9 \\ -0.1 \\ 2.6 \\ -1.7 \\ 10.3 \end{bmatrix}$$

The first output implies that the input sequence has a mean value of $2.1/\sqrt{8}$

The last output has a significantly larger magnitude than all other outputs. It indicates that the input has a strong correlation with this cosine waveform, which implies that the input sequence has a fast rate of change in values.

Discrete Cosine Transform – 8x8 DCT Matrix Example 1

Transposing back to a row, the output sequence is:

$$S[k] = \frac{1}{2}[11.54, 0.10, 0.08, 0.02, -0.11, 0.17, 0.09, 0.13]$$

- Interpret the output:
  - Only the first element is big, all the others are small.
  - This shows that there is a high correlation between the input data and the first row of the transform matrix (the lowest frequency, 0Hz).
  - That is, the input data has little variation.

Discrete Cosine Transform – 8x8 DCT Matrix Example 2

Transposing back to a row, the output sequence is:

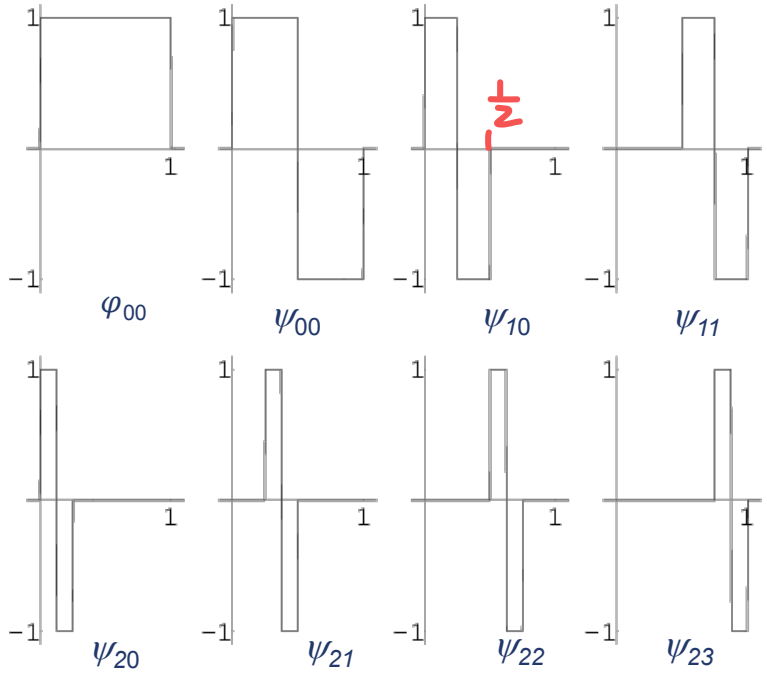
$$S[k] = \frac{1}{2}[6.04, -0.22, 13.68, 1.08, 5.61, -8.27, -0.27, -44.38]$$

- To interpret the output:
  - Only the last element is very big, all the others are relatively small.
  - This shows that there is a high correlation between the input data and the last row of the transform matrix (i.e. the highest frequency).
  - That is, the input data has very large variation. has a fast rate of change in values.

Haar & DWT

H * x'

- Scaled and translated Haar functions



$\varphi_{00} = \varphi(x)$

$\psi_{00} = \psi(x)$

$\psi_{10} = \psi(2x)$

$\psi_{11} = \psi(2x-1)$

$\psi_{20} = \psi(4x)$

$\psi_{21} = \psi(4x-1)$

$\psi_{22} = \psi(4x-2)$

$\psi_{23} = \psi(4x-3)$

Haar Transform Matrix

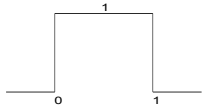
- These two functions can be written as the 2x2 Haar matrix:

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

- All Haar Matrices need to be normalized before applying to signal. The **normalized 2x2 Haar matrix** is

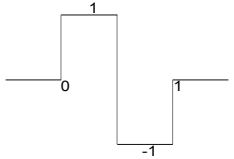
$$H_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Haar **scaling** function



$\varphi_{00} = [1 \quad 1]$

Haar **wavelet** function



$\psi_{00} = [1 \quad -1]$

Haar Transform Matrix

- 4x4 Haar matrix:

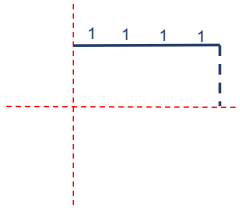
$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix} = \begin{matrix} \varphi_{00} \\ \psi_0 \\ \psi_{1,0} \\ \psi_{1,1} \end{matrix}$$

- The **normalized 4x4 Haar matrix:**

$$H_4 = \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ \sqrt{2} & -\sqrt{2} & 0 & 0 \\ 0 & 0 & \sqrt{2} & -\sqrt{2} \end{bmatrix}$$

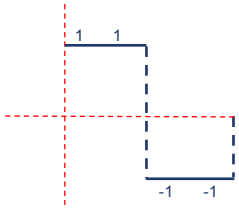
- Normalize each row of the matrix

Haar **scaling** function



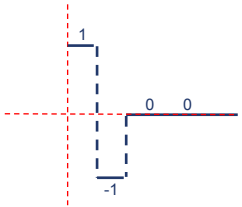
$\varphi_{00} = [1\ 1\ 1\ 1]$

Haar **wavelet** function



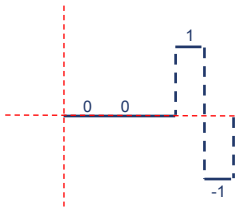
ψ_{00} = [1 1 -1 -1]

Scaled **wavelet** function



ψ_{10} = [1 -1 0 0]

Scaled **wavelet** function



ψ_{11} = [0 0 1 -1]

Haar Transform Matrix

- Unnormalized **8x8 Haar matrix**:

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 \\ 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 \end{bmatrix} = \begin{matrix} \varphi_{00} \\ \psi_{00} \\ \psi_{1,0} \\ \psi_{1,1} \\ \psi_{2,0} \\ \psi_{2,1} \\ \psi_{2,2} \\ \psi_{2,3} \end{matrix}$$

- The 1st row measures the **average** value
- The 2nd row measures a **low frequency component** — slow changing value
- The 3rd and 4th rows measure the **moderate frequency component**
- The 5-8th rows measure the **high frequency component** — narrow shape

Discrete Wavelet Transform – 8x8 Haar Matrix

The un-normalised 8x8 Haar Matrix can be used to show how a Haar Matrix is derived:

$$H_8 = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 \\ 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

$$\begin{matrix} \varphi_0(t) \\ \psi_0(t) \\ \psi_{1,0}(t) \\ \psi_{1,1}(t) \\ \psi_{2,0}(t) \\ \psi_{2,1}(t) \\ \psi_{2,2}(t) \\ \psi_{2,3}(t) \end{matrix}$$

The first row gives the average value of the input sequence. Then subsequent rows correspond to increasing frequencies (similar to DCT). Translation of each frequency allows time information to be determined also.

The matrix would need to be normalised before it could be applied directly to a transform.

Haar Transform Matrix

- Normalized **8x8 Haar matrix**:

$$H_8 = \frac{1}{\sqrt{8}} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ \sqrt{2} & \sqrt{2} & -\sqrt{2} & -\sqrt{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sqrt{2} & \sqrt{2} & -\sqrt{2} & -\sqrt{2} \\ 2 & -2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & -2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & -2 \end{bmatrix}$$

larger ↓ more 0

Matrix only contains real numbers and lots of zeros → low computational complexity

Inverse Haar Transform Matrix

- The Haar transform matrix is real and orthogonal. Thus, the inverse Haar transform can be derived as its transpose:

$$H^{-1} = H^T$$

- You can check this equation by evaluating HH^T, which gives you the identity matrix:

$$HH^T = I$$

- For example, the normalized 4x4 inverse Haar transform matrix is derived as

$$H_4^{-1} = H_4^T = \frac{1}{2}\begin{bmatrix} 1 & 1 & \sqrt{2} & 0 \\ 1 & 1 & -\sqrt{2} & 0 \\ 1 & -1 & 0 & \sqrt{2} \\ 1 & -1 & 0 & -\sqrt{2} \end{bmatrix}$$

Example 5 – Part 1

- Perform Haar Transform on the 8-point input sequence:

$$[\,1, 1, 1, -1, -1, -1, 2, -2]$$

❖ Here is the 8x8 normalized Haar transform matrix:

$$\frac{1}{\sqrt{8}}\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ \sqrt{2} & \sqrt{2} & -\sqrt{2} & -\sqrt{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sqrt{2} & \sqrt{2} & -\sqrt{2} & -\sqrt{2} \\ 2 & -2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & -2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & -2 \end{bmatrix}$$

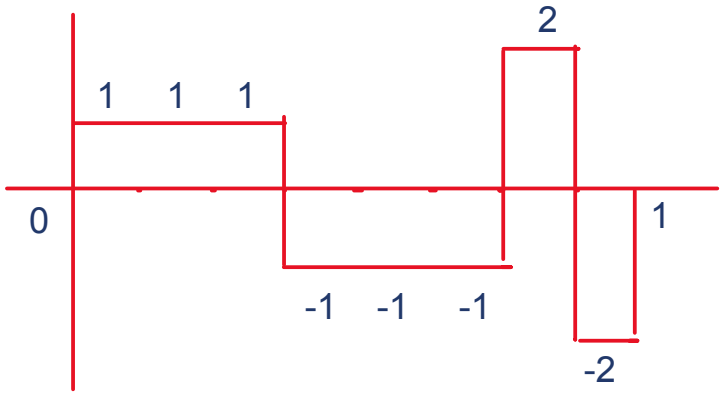
Example 5 – Part 2

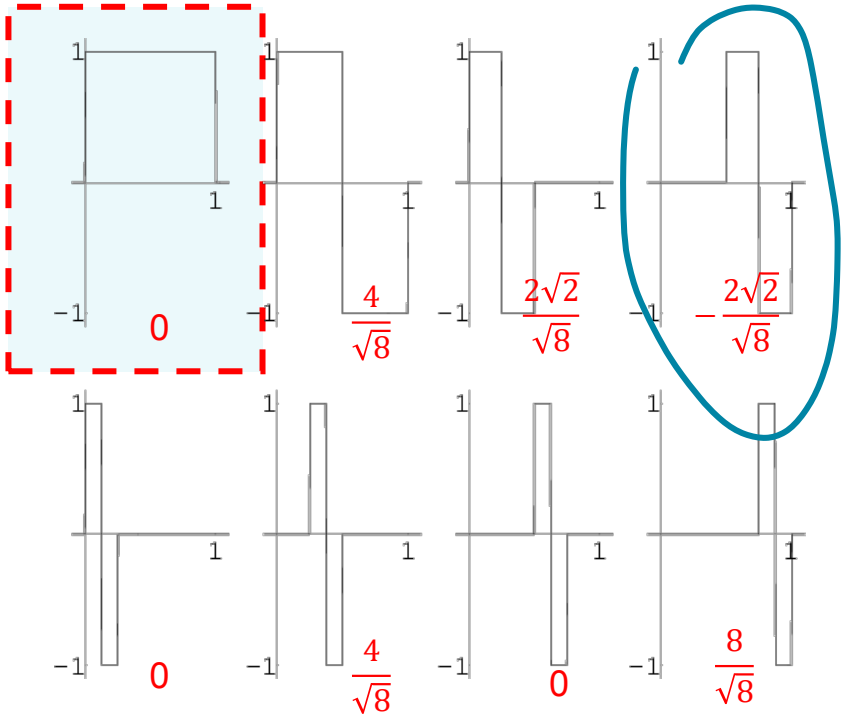
- Given the input sequence and the Haar transform output. Explain the meaning of each transform coefficient in terms of the input
  - From both time and frequency prospective
  - Visualize the input may help

$$\frac{1}{\sqrt{8}}\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ \sqrt{2} & \sqrt{2} & -\sqrt{2} & -\sqrt{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sqrt{2} & \sqrt{2} & -\sqrt{2} & -\sqrt{2} \\ 2 & -2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & -2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & -2 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \\ -1 \\ -1 \\ 2 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{4}{\sqrt{8}} \\ \frac{2\sqrt{2}}{\sqrt{8}} \\ -\frac{2\sqrt{2}}{\sqrt{8}} \\ 0 \\ \frac{4}{\sqrt{8}} \\ 0 \\ \frac{8}{\sqrt{8}} \end{bmatrix}$$

Example 5 – Part 2 Solutions

- The input can be plotted as



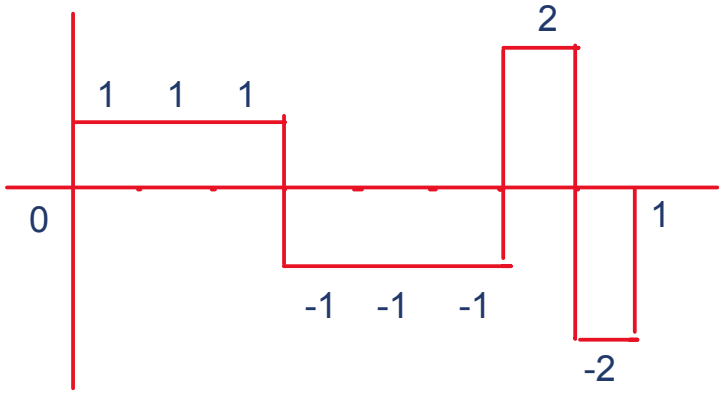


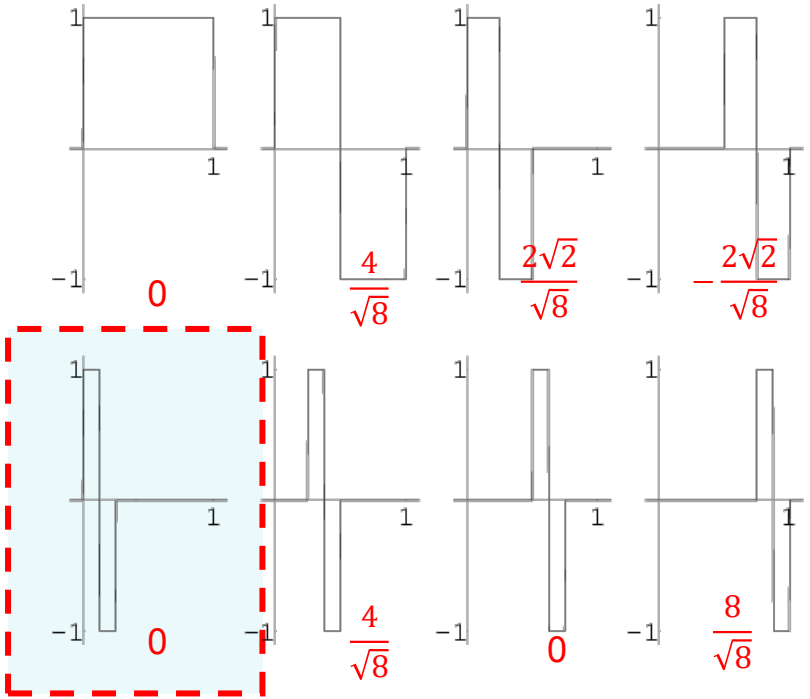
➢ More specifically:

❖ The 1st coef. Indicates that the input has a mean of zero

Example 5 – Part 2 Solutions

- The input can be plotted as



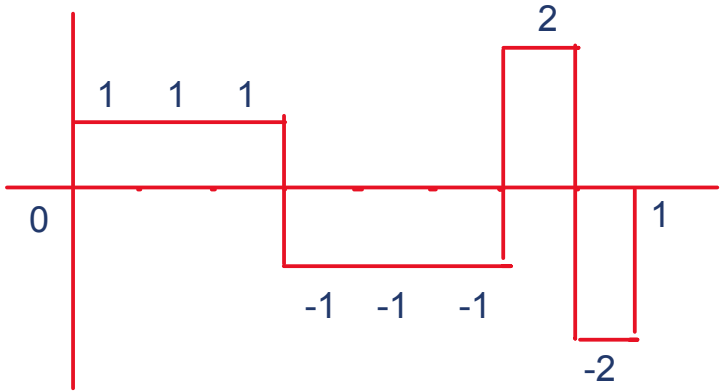


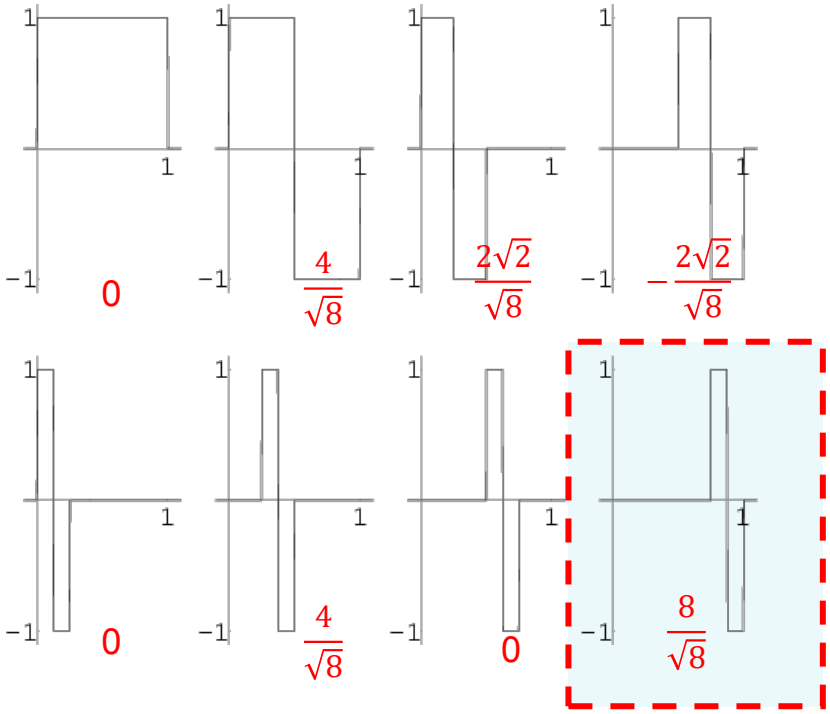
➢ More specifically:

❖ The 5th coef. Indicates that the input has no high-frequency component during $t = [0, \frac{2}{8}]$

Example 5 – Part 2 Solutions

- The input can be plotted as



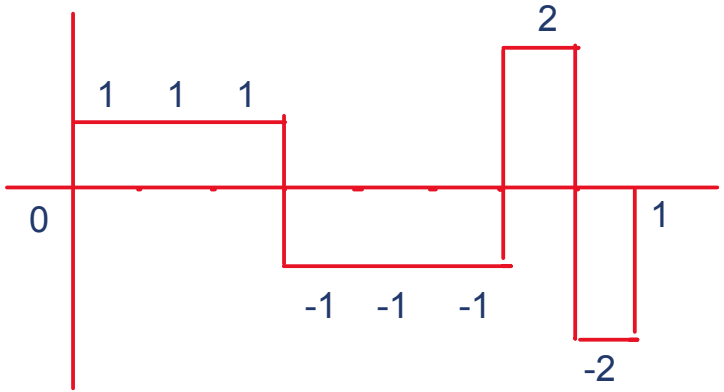


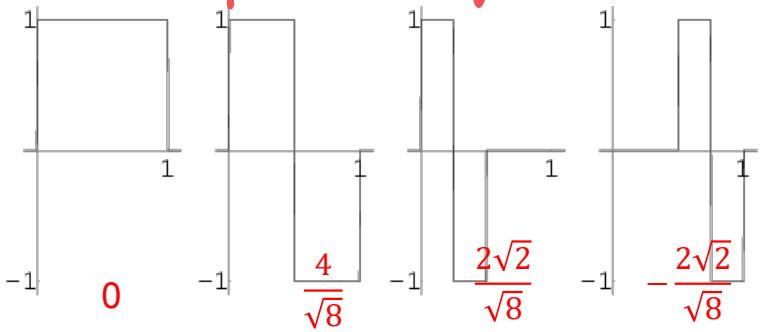
- More specifically:
  - The 8th coef. Indicates that the input has some high-frequency component during time 6-8 $[\frac{6}{8}, \frac{8}{8}]$

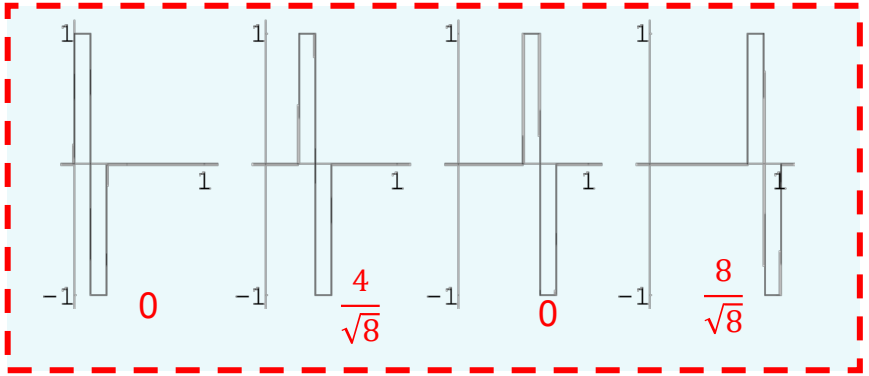
Example 5 – Part 2 Solutions

low magnitude coefficient: signal has no high frequency component during that period. 不重要

- The input can be plotted as







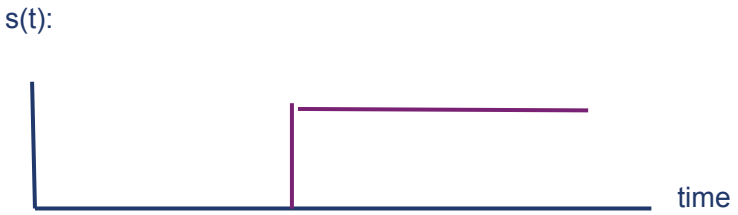
➢ More specifically:

❖ The magnitude of the 8th coef. Is the largest, which indicates the signal has the highest frequency during that period among all time slots

Discrete Wavelet Transform - Tutorial

3. Determine the output of the unnormalised 8x8 Haar Transform for the input sequence:

S[n] = [1, 1, 1, 1, 10, 10, 10, 10]



NOTE: I am using the UNNORMALISED matrix because I am looking for any significant feature in the data. Its exact transform is not required for this. The unnormalised arithmetic is simpler.

Discrete Wavelet Transform - Tutorial

$$H_8 = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 \\ 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 10 \\ 10 \\ 10 \\ 10 \end{bmatrix} = \begin{bmatrix} 44 \\ -36 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$\varphi_0(t)$
$\psi_0(t)$
$\psi_{1,0}(t)$
$\psi_{1,1}(t)$
$\psi_{2,0}(t)$
$\psi_{2,1}(t)$
$\psi_{2,2}(t)$
$\psi_{2,3}(t)$

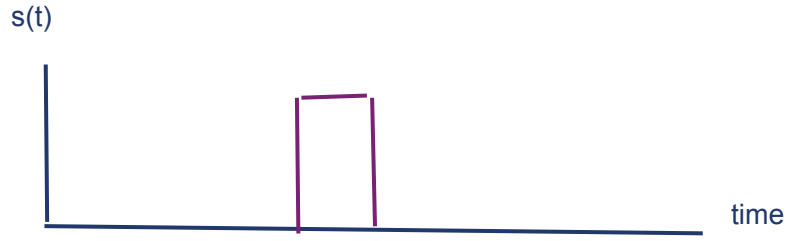
→ in the center

Output indicates a discontinuity corresponding to $\Psi_{0,0}$.
That is, in the centre of the input sequence.

Discrete Wavelet Transform - Tutorial

4. Determine the output of the unnormalised 8x8 Haar Transform for the input sequence:

S[n] = [1, 1, 1, 10, 10, 1, 1, 1]



NOTE: I am using the UNNORMALISED matrix because I am looking for any significant feature in the data. Its exact transform is not required for this. The unnormalised arithmetic is simpler.

Discrete Wavelet Transform - Tutorial

$$H_8 = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 \\ 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 10 \\ 10 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 26 \\ 0 \\ -9 \\ 9 \\ 0 \\ -9 \\ 9 \\ 0 \end{bmatrix}$$

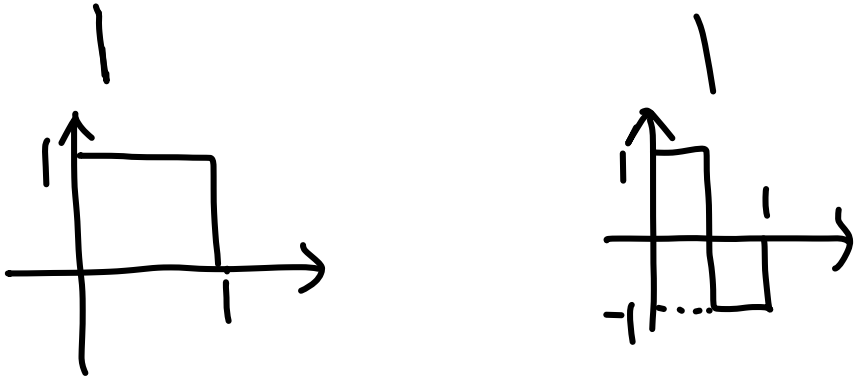
$\varphi_0(t)$
$\psi_0(t)$
$\psi_{1,0}(t)$
$\psi_{1,1}(t)$
ψ_1
$\psi_{2,0}(t)$
$\psi_{2,1}(t)$
$\psi_{2,2}(t)$
$\psi_{2,3}(t)$
ψ_2

Output indicates discontinuities corresponding to shifted Ψ_1 and Ψ_2
That is, identifies a feature.
The resolution would be improved by using a larger Haar Matrix.

Exercise 1

- Using scaled and translated Haar Functions, construct the function given by:

$$1.25\varphi_{0,0}(t)-0.25\varphi_{0,2}(t)-0.25\psi_{0,0}(t)-0.125\psi_{0,2}(t)$$

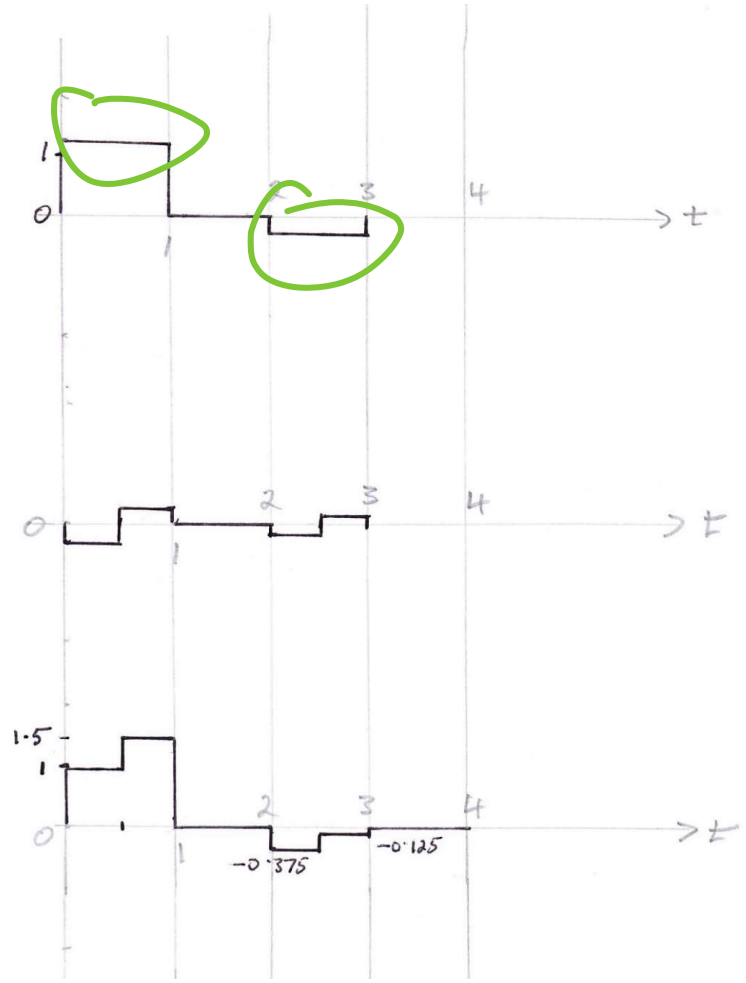


Exercise 1 Solution

$1.25\varphi_{0,0}(t) - 0.25\varphi_{0,2}(t)$

$-0.25\psi_{0,0}(t) - 0.125\psi_{0,2}(t)$

Sum



MRA

Q3

- Multiresolution Analysis (MRA) is used to separate data into coarse and fine detail.
- Apply the transform defined by
- $x_{n-1},i = (x_{n,2i}+x_{n,2i+1})/2$
- $d_{n-1},i = (x_{n,2i}-x_{n,2i+1})/2$
- to the sequence
- $[x_{n,i}] = [12, 8, 13, 10, 6, 7, 11, 14]$
- Where i = 0…..7, is the index position in the sequence, and
- n is the level. The next level is n-1.
- At each level, concatenate the sequencies for x_{n-1},i and d_{n-1},i
- Continue till no further levels are possible.
- i) State the significance of the first element in the final level.
- ii) Has any information been lost in the process?
- iii) Comment on how this process could be used to compress the data.

Q3 Answer

Applying the transform:

- n=3 [12, 8, 13, 10, 6, 7, 11, 14]
- n=2 [10, 11.5, 6.5, 12.5, 2, 1.5, -0.5, -1.5]
- n=1 [10.75, 9.5, -0.75, -3, 2, 1.5, -0.5, -1.5]
- n=0 [10.125, 0.625, -0.75, -3, 2, 1.5, -0.5, -1.5]
- i) 4 levels
- ii) The first element is the average of all the elements in the original sequence.
- iii) The remaining 7 elements are the fine detail in the original data.
- iv) No information has been lost
- v) Because most of the values in the final level are small, potentially fewer bits would be required to store it. Where there are zeroes, they do not need to be stored, although the positions of the other values would need to be stored. Small values could be replaced by zeroes without significant loss of detail, this can be done by applying a threshold value, the bigger the threshold the greater the loss of detail.

vi) $X_{n,2i} = X_{n-1,i} + d_{n-1,i}$

$X_{n,2i+1} = X_{n-1,i} - d_{n-1,i}$

those 2 equations can be used to reconstruct the original data from the final level.

vii) The Haar DWT is used to perform averaging and differentiating, the coarse values by the Haar scaling function and the fine details by the Haar Wavelet function.

WTFB

Q5

Use the recursive equations:

$$c_{m-1,n} = \sqrt{2} \cdot \frac{1}{2}(c_{m,2n} + c_{m,2n+1})$$
$$= \frac{1}{\sqrt{2}}(c_{m,2n} + c_{m,2n+1})$$
$$d_{m-1,n} = \frac{1}{\sqrt{2}}(c_{m,2n} - c_{m,2n+1})$$

Suppose we have a Haar wavelet transform analysis filterbank which uses a low-pass filter

$$h_0[0] = h_0[1] = \frac{1}{2}$$

and a high-pass filter $h_1[0] = \frac{1}{2}$ $h_1[1] = -\frac{1}{2}$

Using the recursive equations, calculate the Haar wavelet transform for a sampled signal

$$s[n] = [1, 2, 0, -2]$$

after 1 and 2 stages of the transform filterbank.

Calculate the inverse wavelet transform of your result using the resynthesis filterbank, to confirm that the result of the inverse transform is the original sequence.

Q5 Answer

Input sequence: $[1, 2, 0, -2]$

$$c_{1,0} = \frac{1}{\sqrt{2}}(1 + 2) = 3 / \sqrt{2}$$

$$c_{1,1} = \frac{1}{\sqrt{2}}(0 + (-2)) = -2 / \sqrt{2}$$

$$d_{1,0} = \frac{1}{\sqrt{2}}(1 - 2) = -1 / \sqrt{2}$$

$$d_{1,1} = \frac{1}{\sqrt{2}}(0 - (-2)) = 2 / \sqrt{2}$$

After 1 level: $\frac{1}{\sqrt{2}}(3, -2, -1, 2)$

$$c_{0,0} = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}(3 + (-2)) = 1/2$$

$$d_{0,0} = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}(3 - (-2)) = 5/2$$

After 2 levels: $\frac{1}{2}(1, 5, -\sqrt{2}, 2\sqrt{2})$

Q5 Answer 2: Inverse

From level 0 to level 1:

$$c_{1,0} = \sqrt{2}\left(\tfrac{1}{2}c_{0,0} + \tfrac{1}{2}d_{0,0}\right) = \sqrt{2}\left(\tfrac{1}{2}1/2 + \tfrac{1}{2}5/2\right) = \sqrt{2}(6/4) = 3/\sqrt{2}$$

$$c_{1,1} = \sqrt{2}\left(\tfrac{1}{2}c_{0,0} - \tfrac{1}{2}d_{0,0}\right) = \sqrt{2}\left(\tfrac{1}{2}1/2 - \tfrac{1}{2}5/2\right) = \sqrt{2}(-4/4) = -2/\sqrt{2}$$

The *d* coefficients in the 2nd half of the WT are unchanged, so we have

$$\tfrac{1}{\sqrt{2}}(3, -2, -1, 2)$$

which is the same as the first stage transform in the analysis direction.

From level 1 to level 2:

$$c_{2,0} = \sqrt{2}\left(\tfrac{1}{2}c_{1,0} + \tfrac{1}{2}d_{1,0}\right) = \sqrt{2}\left(\tfrac{1}{2}3/\sqrt{2} + \tfrac{1}{2}(-1/\sqrt{2})\right) = \frac{1}{2}(3-1) = 1$$

$$c_{2,1} = \sqrt{2}\left(\tfrac{1}{2}c_{1,0} - \tfrac{1}{2}d_{1,0}\right) = \sqrt{2}\left(\tfrac{1}{2}3/\sqrt{2} - \tfrac{1}{2}(-1/\sqrt{2})\right) = \frac{1}{2}(3+1) = 2$$

$$c_{2,2} = \sqrt{2}\left(\tfrac{1}{2}c_{1,1} + \tfrac{1}{2}d_{1,1}\right) = \sqrt{2}\left(\tfrac{1}{2}.-2/\sqrt{2} + \tfrac{1}{2}2/\sqrt{2}\right) = \frac{1}{2}(-2+2) = 0$$

$$c_{2,3} = \sqrt{2}\left(\tfrac{1}{2}c_{1,1} - \tfrac{1}{2}d_{1,1}\right) = \sqrt{2}\left(\tfrac{1}{2}.-2/\sqrt{2} - \tfrac{1}{2}2/\sqrt{2}\right) = \frac{1}{2}(-2-2) = -2$$

$$[1, 2, 0, -2]$$

KLT – Procedures



$$\mathbf{X} = [\vec{x}_0, \vec{x}_1, \ldots \vec{x}_{N-1}]$$

1. Find mean vector for input data $E(\mathbf{X}) = \frac{1}{N}\sum_{i=0}^{N-1} \vec{x}_i$ mean(x) mean(y)
2. Find covariance matrix $\mathbf{R_{XX}} = \frac{1}{N-1}\sum_{i=0}^{N-1} (\vec{x}_i - E(\vec{x}))(\vec{x}_i - E(\vec{x}))^T$ cov(x,y) =A
3. Find eigenvalues of the covariance matrix $|\mathbf{R_{XX}} - \lambda \mathbf{I}| = 0$
4. Find eigenvectors of the covariance matrix $(\mathbf{R_{XX}} - \lambda_i \mathbf{I})\vec{\varphi}_i = 0$

} [V,D]=eig(A)

V: 特征向量 (列)
D: の (对角)

5. Normalise the eigenvectors $\vec{\varphi}^*{}_i = \frac{\vec{\varphi}_i}{|\vec{\varphi}_i|}$ so that $\langle \vec{\varphi}_i, \vec{\varphi}_i \rangle = 1$
6. Transform the input $\mathbf{Y} = \varphi^T \mathbf{X}$, where $\varphi^T = [\vec{\varphi}^*{}_1, \vec{\varphi}^*{}_2, \ldots]$ V'*X
7. Set the last row of the transformed data to 0
8. Invert the transform

KLT - Example

mean(x,y)
covariance matrix = $\begin{bmatrix} \text{var}(x) & \text{cov}_{xy} \\ \text{cov}_{xy} & \text{var}_y \end{bmatrix}$
$\text{cov}(x,y)$

- Determine the KLT of the following 2D data set.

$$\mathbf{X} = \begin{bmatrix} 2 & 4 & 5 & 5 & 3 & 2 \\ 2 & 3 & 4 & 5 & 4 & 3 \end{bmatrix} \begin{matrix} x \\ y \end{matrix}$$

(Assume this is a sample of a larger population)
(Each row represents a feature/variable)

- Then, replace the elements of the least principal component of the output by zero and then perform the inverse KLT

KLT - Example

$$\mathbf{X} = \begin{bmatrix} 2 & 4 & 5 & 5 & 3 & 2 \\ 2 & 3 & 4 & 5 & 4 & 3 \end{bmatrix}$$

i sample; 6th sample

*Each row corresponds to a variable

1. Mean vector for input data

$$E(\vec{x}) = \frac{1}{6}\begin{bmatrix} 2+4+5+5+3+2 \\ 2+3+4+5+4+3 \end{bmatrix} = \begin{bmatrix} 3.5 \\ 3.5 \end{bmatrix}$$

2. Covariance matrix

$$\mathbf{R}_{xx} = \frac{1}{5}\sum_{i=0}^{5}(\vec{x}_i - \begin{bmatrix} 3.5 \\ 3.5 \end{bmatrix})(\vec{x}_i - \begin{bmatrix} 3.5 \\ 3.5 \end{bmatrix})^T$$

0 to N-1; N-1

$$= \frac{1}{5}\left\{\begin{bmatrix} -1.5 \\ -1.5 \end{bmatrix}\begin{bmatrix} -1.5 & -1.5 \end{bmatrix} + \ldots + \begin{bmatrix} -1.5 \\ -.5 \end{bmatrix}\begin{bmatrix} -1.5 & -.5 \end{bmatrix}\right\}$$

Transpoze

$$= \frac{1}{5}\left\{\begin{bmatrix} 2.25 & 2.25 \\ 2.25 & 2.25 \end{bmatrix} + \ldots + \begin{bmatrix} 2.25 & 0.75 \\ 0.75 & 0.25 \end{bmatrix}\right\} = \begin{bmatrix} 1.9 & 1.1 \\ 1.1 & 1.1 \end{bmatrix}$$

KLT - Example

$$\mathbf{X} = \begin{bmatrix} 2 & 4 & 5 & 5 & 3 & 2 \\ 2 & 3 & 4 & 5 & 4 & 3 \end{bmatrix}$$

*Each row corresponds to a variable

3. Find eigenvalues

$$0 = |\mathbf{R}_{xx} - \lambda \mathbf{I}| = \begin{vmatrix} 1.9 - \lambda & 1.1 \\ 1.1 & 1.1 - \lambda \end{vmatrix}$$

$$\lambda^2 - 3\lambda + 0.88 = 0$$

$$\lambda_1 = 2.67, \lambda_2 = 0.33$$

4. Find eigenvectors

$$\begin{bmatrix} -0.77 & 1.1 \\ 1.1 & -1.57 \end{bmatrix} \begin{bmatrix} \varphi_{11} \\ \varphi_{21} \end{bmatrix} = 0 \qquad \begin{bmatrix} 1.57 & 1.1 \\ 1.1 & 0.77 \end{bmatrix} \begin{bmatrix} \varphi_{12} \\ \varphi_{22} \end{bmatrix} = 0$$

$$\Rightarrow \varphi_{11} = 1.43\varphi_{21} \qquad \Rightarrow \varphi_{12} = -0.7\varphi_{22}$$

KLT - Example

$$\mathbf{X} = \begin{bmatrix} 2 & 4 & 5 & 5 & 3 & 2 \\ 2 & 3 & 4 & 5 & 4 & 3 \end{bmatrix}$$

*Each row corresponds to a variable

5. Normalise and solve (From previous step: $\varphi_{11} = 1.43\varphi_{21},\ \varphi_{12} = -0.7\varphi_{22}$)

$$\langle \vec{\varphi}_1, \vec{\varphi}_1 \rangle = 1 \rightarrow {\varphi_{11}}^2 + {\varphi_{21}}^2 = 1$$
$$\langle \vec{\varphi}_2, \vec{\varphi}_2 \rangle = 1 \rightarrow {\varphi_{12}}^2 + {\varphi_{22}}^2 = 1$$

$$\Rightarrow \varphi = \begin{bmatrix} \varphi_{11} & \varphi_{12} \\ \varphi_{21} & \varphi_{22} \end{bmatrix} = \begin{bmatrix} 0.82 & -0.57 \\ 0.57 & 0.82 \end{bmatrix}$$

6. Transform the input

$$\mathbf{Y} = \varphi^T \mathbf{X} = \begin{bmatrix} 0.82 & -0.57 \\ -0.57 & 0.82 \end{bmatrix} \begin{bmatrix} 2 & 4 & 5 & 5 & 3 & 2 \\ 2 & 3 & 4 & 5 & 4 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 2.78 & 4.99 & 6.38 & 6.95 & 4.74 & 3.35 \\ 0.5 & 0.18 & 0.43 & 1.25 & 1.57 & 1.32 \end{bmatrix}$$

KLT - Example

$$\mathbf{X} = \begin{bmatrix} 2 & 4 & 5 & 5 & 3 & 2 \\ 2 & 3 & 4 & 5 & 4 & 3 \end{bmatrix}$$

Original data

$$\mathbf{Y} = \begin{bmatrix} 2.78 & 4.99 & 6.38 & 6.95 & 4.74 & 3.35 \\ 0.5 & 0.18 & 0.43 & 1.25 & 1.57 & 1.32 \end{bmatrix}$$

Transformed data
(projection onto orthogonal space)

have 0 covariance

two "0"

Check: Covariance of Transformed Data

$$\mathbf{R_{YY}} = \frac{1}{N-1}\sum_{i=0}^{N-1}(\vec{y}_i - E[\vec{y}])(\vec{y}_i - E[\vec{y}])^T = \begin{bmatrix} 2.67 & 0 \\ 0 & 0.33 \end{bmatrix}$$

Total "energy" is proportional to 2.67+0.33 = 3.00

- 89% (2.67/3 *100%) of energy along the first axis, 11% on the second.
- Covariance between two new variables is zero, i.e., uncorrelated.

KLT - Example

$$\mathbf{X} = \begin{bmatrix} 2 & 4 & 5 & 5 & 3 & 2 \\ 2 & 3 & 4 & 5 & 4 & 3 \end{bmatrix}$$

Original data

$$\mathbf{Y} = \begin{bmatrix} 2.78 & 4.99 & 6.38 & 6.95 & 4.74 & 3.35 \\ 0.5 & 0.18 & 0.43 & 1.25 & 1.57 & 1.32 \end{bmatrix}$$

Transformed data
(projection onto orthogonal space)

Now suppose we do dimensionality reduction and remove second coordinate.

$$\mathbf{Y}' = \begin{bmatrix} 2.78 & 4.99 & 6.38 & 6.95 & 4.74 & 3.35 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

KLT - Example

$$\mathbf{X} = \begin{bmatrix} 2 & 4 & 5 & 5 & 3 & 2 \\ 2 & 3 & 4 & 5 & 4 & 3 \end{bmatrix}$$

Original data

- Inverse transform of reduced data

compressed data

$$\mathbf{X}' = \varphi \mathbf{Y}' = \begin{bmatrix} 0.82 & -0.57 \\ 0.57 & 0.82 \end{bmatrix} \begin{bmatrix} 2.78 & 4.99 & 6.38 & 6.95 & 4.74 & 3.35 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2.28 & 4.1 & 5.23 & 5.7 & 3.89 & 2.75 \\ 1.58 & 2.84 & 3.64 & 3.96 & 2.7 & 1.91 \end{bmatrix}$$

if use 1 principle component

- Error

$$J = E(|\mathbf{X} - \mathbf{X}'|^2)$$

This MSE could be calculated, but without knowing the context of the application of the initial data would be meaningless.

Karhunen Loève Transform (KLT) – Tutorial Solution

$$X = \begin{bmatrix} -1 & -2 & 0 & 0 & 2 \\ 0 & -1 & 2 & -1 & 4 \\ 2 & 4 & 0 & -2 & 0 \end{bmatrix}$$

a	b	c
-1	0	2
-2	-1	4
0	2	0
0	-1	-2
2	4	0

- Step 1: Find the mean vector

$$E(X) = \begin{bmatrix} -0.2 \\ 0.8 \\ 0.8 \end{bmatrix}$$

- Step 2: Find covariance matrix

In Question 1, we have $var_a = 2, var_b = 4.7, cov_{a,b} = 2.7$

Only need to compute $var_c =$, $cov_{a,c} =$, $cov_{b,c} =$

$$var_c = \frac{1}{N-1}\sum_{0}^{N-1}(c_i - c_{mean})^2 = \frac{1}{4}[(2-0.8)^2 + (4-0.8)^2 + 0.8^2 + (-2-0.8)^2 + 0.8^2] = 5.2$$

Karhunen Loève Transform (KLT) – Tutorial Solution

$$X = \begin{bmatrix} -1 & -2 & 0 & 0 & 2 \\ 0 & -1 & 2 & -1 & 4 \\ 2 & 4 & 0 & -2 & 0 \end{bmatrix}$$

a	b	c
-1	0	2
-2	-1	4
0	2	0
0	-1	-2
2	4	0

- Step 1: Find the mean vector

$$E(X) = \begin{bmatrix} -0.2 \\ 0.8 \\ 0.8 \end{bmatrix}$$

- Step 2: Find covariance matrix

In Question 1, we have $var_a = 2, var_b = 4.7, cov_{a,b} = 2.7$

Only need to compute $var_c = 5.2,\ cov_{a,c} =$, $cov_{b,c} =$

$$cov_{a,c} = \frac{1}{N-1}\sum_{0}^{N-1}(a_i - a_{mean})(c_i - c_{mean})$$

$$= \frac{1}{4}[(-1+0.2)(2-0.8) + (-2+0.2)(4-0.8) + 0.2(-0.8) + 0.2(-2-0.8) + (2+0.2)(-0.8)]$$

$$= -2.3$$

Karhunen Loève Transform (KLT) – Tutorial Solution

$$X = \begin{bmatrix} -1 & -2 & 0 & 0 & 2 \\ 0 & -1 & 2 & -1 & 4 \\ 2 & 4 & 0 & -2 & 0 \end{bmatrix}$$

a	*b*	*c*
-1	0	2
-2	-1	4
0	2	0
0	-1	-2
2	4	0

- Step 1: Find the mean vector

$$E(X) = \begin{bmatrix} -0.2 \\ 0.8 \\ 0.8 \end{bmatrix}$$

- Step 2: Find covariance matrix

In Question 1, we have $var_a = 2, var_b = 4.7, cov_{a,b} = 2.7$

Only need to compute $var_c = 5.2,\ cov_{a,c} = -2.3,\ cov_{b,c} =$

$$cov_{b,c} = \frac{1}{N-1}\sum_{0}^{N-1}(b_i - b_{mean})(c_i - c_{mean})$$

$$= \frac{1}{4}[(-0.8)(2-0.8) + (-1-0.8)(4-0.8) + (2-0.8)(-0.8) + (-1-0.8)(-2-0.8) + (4-0.8)(-0.8)]$$

$$= -1.3$$

Karhunen Loève Transform (KLT) – Tutorial Solution

$$X = \begin{bmatrix} -1 & -2 & 0 & 0 & 2 \\ 0 & -1 & 2 & -1 & 4 \\ 2 & 4 & 0 & -2 & 0 \end{bmatrix}$$

a	b	c
-1	0	2
-2	-1	4
0	2	0
0	-1	-2
2	4	0

- Step 1: Find the mean vector

$$E(X) = \begin{bmatrix} -0.2 \\ 0.8 \\ 0.8 \end{bmatrix}$$

- Step 2: Find covariance matrix

In Question 1, we have $var_a = 2, var_b = 4.7, cov_{a,b} = 2.7$

We just calculated $var_c = 5.2,\ cov_{a,c} = -2.3,\ cov_{b,c} = -1.3$

Hence, the covariance matrix is

$$R_{a,b,c} = \begin{bmatrix} 2 & 2.7 & -2.3 \\ 2.7 & 4.7 & -1.3 \\ -2.3 & -1.3 & 5.2 \end{bmatrix}$$

Transform Method

Transforms are used for a variety of purposes. For example:

- **Fourier Transforms** are used to obtain the frequency spectrum of a function of time.
- **Cosine Transforms** are used to identify the rate-of-change of the input data.
- **Wavelet Transforms** are used to identify trends in the input data ("approximations") and to identify short-duration features or artifacts in the input data.

variation

where the change occurs in the signal

Transform Methods – Exercise 1

For a 4-point input sequence

$$S[n] = [6, 3, -2.5, 7]$$

Determine the output from each of the following discrete transforms:

a. DFT

b. DCT

c. DWT, using a Haar wavelet function.

Transform Methods – Exercise 1 Solution

$N=4$

$$\text{DFT} = \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix}\begin{bmatrix} 6 \\ 3 \\ -2.5 \\ 7 \end{bmatrix} = \begin{bmatrix} 6.75 \\ 4.25 + 2i \\ -3.25 \\ 4.25 - 2i \end{bmatrix}$$

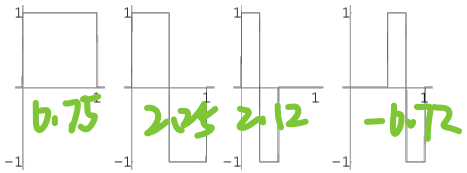
Transform Methods – Exercise 1 Solution

$$\text{DCT} = \begin{array}{c} \\ k=0 \\ k=1 \\ k=2 \\ k=3 \end{array} \begin{array}{c} \begin{array}{cccc} n=0 & n=1 & n=2 & n=3 \end{array} \\ \begin{bmatrix} 0.50 & 0.50 & 0.50 & 0.50 \\ 0.65 & 0.27 & -0.27 & -0.65 \\ 0.50 & -0.50 & -0.50 & 0.50 \\ 0.27 & -0.65 & 0.65 & -0.27 \end{bmatrix} \end{array} \begin{bmatrix} 6 \\ 3 \\ -2.5 \\ 7 \end{bmatrix} = \begin{bmatrix} 6.75 \\ 0.835 \\ 6.25 \\ -3.845 \end{bmatrix}$$

Transform Methods – Exercise 1 Solution

$$\text{DHT} = \frac{1}{2}\underbrace{\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ \sqrt{2} & -\sqrt{2} & 0 & 0 \\ 0 & 0 & \sqrt{2} & -\sqrt{2} \end{bmatrix}}_{H_4} \begin{bmatrix} 6 \\ 3 \\ -2.5 \\ 7 \end{bmatrix} = \begin{bmatrix} 6.75 \\ 2.25 \\ 2.12 \\ -6.72 \end{bmatrix}$$

Transform Methods – Exercise 1 Solution



Input: S[n] = [6, 3, -2.5, 7]

Same first output

$$\text{DFT} = \begin{bmatrix} 6.75 \\ 4.25 + 2i \\ -3.25 \\ 4.25 - 2i \end{bmatrix} \quad \text{DCT} = \begin{bmatrix} 6.75 \\ 0.835 \\ 6.25 \\ -3.845 \end{bmatrix} \quad \text{DHT} = \begin{bmatrix} 6.75 \\ 2.25 \\ 2.12 \\ -6.72 \end{bmatrix}$$



strong correlation to

large

large

$\frac{6.75}{\sqrt{N}}$

- The first output of all three methods are the same. i.e. the smoothed value, divide by $\sqrt{N}$ to obtain mean
- DFT is the only one that produces complex outputs

has a fast changing value at last 2 time slot.

Sharp increase in the input (∵ negative)

- Three transform methods are designed for different purposes
  - DFT – frequency spectrum as a function of time
  - DCT – rate of change in the input
  - DHT – identify short duration features

Daubechies Wavelet

The forward transform matrix for the dbk-2 transform is:

$$\begin{bmatrix} h_0[0] & h_0[1] & h_0[2] & h_0[3] & 0 & 0 & 0 & 0 \\ h_1[0] & h_1[1] & h_1[2] & h_1[3] & 0 & 0 & 0 & 0 \\ 0 & 0 & h_0[0] & h_0[1] & h_0[2] & h_0[3] & 0 & 0 \\ 0 & 0 & h_1[0] & h_1[1] & h_1[2] & h_1[3] & 0 & 0 \\ 0 & 0 & 0 & 0 & h_0[0] & h_0[1] & h_0[2] & h_0[3] \\ 0 & 0 & 0 & 0 & h_1[0] & h_1[1] & h_1[2] & h_1[3] \\ 0 & 0 & 0 & 0 & 0 & 0 & h_0[0] & h_0[1] & h_0[2] & h_0[3] \\ 0 & 0 & 0 & 0 & 0 & 0 & h_1[0] & h_1[1] & h_1[2] & h_1[3] \end{bmatrix} \begin{bmatrix} s_0 \\ s_1 \\ s_2 \\ s_3 \\ s_4 \\ s_5 \\ s_6 \\ s_7 \end{bmatrix} = \begin{bmatrix} c_1 \\ d_1 \\ c_2 \\ d_2 \\ c_3 \\ d_3 \\ c_4 \\ d_4 \end{bmatrix}$$

- $h_0[n]=[h_0[0], h_0[1], h_0[2], h_0[3]] =[0.483, 0.837, 0.224, -0.129]$
- ✓ $h_1[n]=[h_0[3],- h_0[2], h_0[1], -h_0[0]] = [-0.129, -0.224, 0.837, -0.483]$
- ✓ $y_0[n]=[h_0[3], h_0[2], h_0[1], h_0[0]] = [-0.129, 0.224, 0.837, 0.483]$
- ✓ $y_1[n]=[-h_0[0], h_0[1], -h_0[2], h_0[3]] = [-0.483, 0.837, -0.224, -0.129]$

Calculate the first 2 c coefficients and the first 2 d coefficients if the input sequence is:
S[n] = [3, 7, 1, 4, 6, 9, 2, 5]

Daubechies Wavelet – Exercise Solution

- c_1 = 3x0.483 + 7x0.837 + 1x0.224 + 4x(-0.129) = 7.016
- d_1= 3x(-0.129) + 7x(-0.224) + 1x0.837 + 4x(-0.483) = -3.05
- c_2 = 1x0.483 + 4x0.837 + 6x0.224 + 9x(-0.129) = 4.014
- d_2 = 1x(-0.129) + 4x(-0.224) + 6x0.837 + 9x(-0.483) = -0.35